Does intelligence compound when each step builds on the last, or merely accumulate?

Theory · Theory paper · Paper I

Recursive Dynamics · The ARC Theory · Law I, the ARC Principle, a law under test

Michael Darius Eastwood

Independent Researcher, London, United Kingdom

ORCID 0009-0004-3222-7442 michael@michaeldariuseastwood.com

Abstract

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. Does intelligence compound when each step builds on the last, or merely accumulate? This paper formalises Law I of the ARC Theory, the ARC Principle (Artificial Recursive Creation): usable capability $U = IR^{\alpha}$, with its exponent measured rather than assumed. Its stronger hypothesis is that sequential recursion scales super-linearly and faster than parallel sampling. Published reasoning-model data give error-decay slopes of about 0.1 to 0.3 for parallel sampling and 1.34 from two sequential points. These compare error against sample count or average tokens, not directly capability against recursive depth; the series also differ in model and benchmark, and the sequential points are two model versions. Under a positive-error power model, the proxy $1/E$ has the same exponent on the same resource axis. Reading it as the capability exponent requires an explicit metric and depth mapping. The later five-model study, Paper II, has one interpretable sequential fit, on Gemini 3 Flash: error-decay regression 0.49 (standard error 0.20) and endpoint estimate 0.59 (bootstrap 95 per cent interval −1.3 to 2.9). That one-model result is inconclusive. Neither set independently measures self-referential coupling. The paper states the ARC Bound, the conjectured value two of Law III's stability ceiling, with its conditions and falsification criteria. Super-linear capability growth remains unresolved; Law I remains a law under test.

Version 1.18 distinguishes error decay, capability growth and coupling; Appendix F records the correction of 1 November 2026 and preserves the earlier wording. Appendix B keeps the preceding version history.

Paper I of the ARC Theory, within Recursive Dynamics: it states Law I, the ARC Principle, formally as the ARC Equation and examines error-decay evidence under an explicit metric bridge, and it states the ARC Bound, the conjectured value of Law III, the ARC Ceiling, each a law under test; it uses Law II's criterion to say what stable means, and it decides none of P15, P12 and P21, the propositions of Laws IV and V (registered as laws on 15 October 2026 in version 1.103 of the theory registration; Section 6.3 gives their status). Working paper, version 1.18, 1 November 2026; first published 17 January 2026; Appendices B and F list every version and correction. Recorded on OSF as *Paper I · The ARC Theory · Law I, the ARC Principle · The ARC Equation, the Law of Conversion*, DOI 10.17605/OSF.IO/B6N27. About the author, founder of Recursive Dynamics, without an institution or a doctorate: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; his book, *Infinite Architects* (first edition, 2 January 2026): osf.io/dqm3e. Licence: CC BY 4.0.

Contents

1 Introduction **4**

2 Related work, antecedents and the rival instrument **5**

3 The ARC Principle and its framework **8**
3.1 The ARC Equation . . . 8
3.2 Recursion and its two forms . . . 9
3.3 The quadratic stability ceiling hypothesis . . . 10

4 Empirical analysis **11**
4.1 Data sources . . . 11
4.2 Method . . . 11
4.3 Parallel recursion . . . 11
4.4 Sequential recursion . . . 12
4.5 Summary of findings . . . 12

5 Falsification **13**
5.1 Criteria and their status . . . 13
5.2 What would refute the principle . . . 14
5.3 The decision form . . . 14
5.4 An objection pre-empted . . . 14
5.5 A programme hypothesis beyond this paper . . . 15

6 What this paper does and does not show **15**
6.1 The claims and their status . . . 15
6.2 The exponent-composition identity . . . 16
6.3 Laws IV and V, and the programme's status . . . 17

7 Limitations **17**

8 Implications **18**
8.1 For AI development . . . 18
8.2 For AI safety . . . 18
8.3 For scientific understanding . . . 18

9 Conclusion **19**

References **20**

A Proofs **25**

B Version history and corrections **25**
B.1 Version 1.17 . . . 25
B.2 Earlier versions, as printed . . . 27
B.3 Notes moved from the body, as printed . . . 29
B.4 Earlier titles . . . 30
B.5 Where the layout of version 1.16 now stands . . . 30

C Priority record **31**
C.1 Dated records . . . 31
C.2 Dated predictions and this paper, as recorded on 25 August 2026 . . . 32

D Notation and glossary **33**
D.1 Symbols . . . 33
D.2 Earlier senses of two letters . . . 33
D.3 Glossary . . . 34

E The paper's place in the ARC Theory **34**
E.1 The five laws and this paper . . . 34
E.2 Its position . . . 35

F Metric clarification of 1 November 2026 and preserved wording **35**
F.1 Version 1.18: what changed and why . . . 35
F.2 Earlier wording, grouped in the correction order . . . 35
F.3 Earlier figure, retained as historical evidence . . . 41

1 Introduction

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. If control of a capable AI system were lost and could not be won back, what the system itself carries would decide what it does. The programme's position is that people keep oversight, containment and shutdown for as long as they physically can, and never give them up by choice; correction built into a system is the contingency for control lost against our will, not a reason to let go of it. Whether any correction can keep pace depends first on how capability grows when a system works on its own output, and that is this paper's question: does intelligence compound when each step builds on the last, or merely accumulate? In plain words, its answer so far is this. Published reasoning-model data give steeper error reduction against average tokens in one sequential pair than against sample count in a parallel series. These are different response-to-resource comparisons, with other differences between the systems; they do not isolate the form of reasoning or directly measure capability against recursive depth. The one later interpretable error-decay fit is inconclusive. The question stays open, and the paper states it in a form that can be tested and refuted.

This paper belongs to the ARC Theory (the Theory of Artificial Recursive Creation), the founding theory of Recursive Dynamics, the science of systems that improve themselves, which the author founded (the founding paper). The theory is carried by five laws, each a law under test: Law I, the ARC Principle, that usable capability is a power of recursive depth, with the exponent measured rather than assumed; Law II, the ARC Co-Scaling Law; Law III, the ARC Ceiling; and Laws IV and V, the ARC Persistence Law and the ARC Embedding Law, added on 24 September 2026 and registered as laws on 15 October 2026, their mathematics in development. The laws and twenty-two propositions are registered on OSF (the theory registration), and the statement paper states the theory whole. This paper states Law I formally and examines error-decay evidence under the metric and depth premises stated in Section 4.2, and it states the ARC Bound, the conjectured value of Law III's ceiling; Appendix E places it among the laws.

The scaling laws governing artificial intelligence have been extensively studied. Kaplan et al. (2020) established power-law relationships between model performance and parameters, while Hoffmann et al. (2022) refined these with compute-optimal training prescriptions. These laws govern *what* to scale but do not address *why* scaling produces intelligent behaviour.

The emergence of reasoning models in 2024 and 2025 introduced a new variable: test-time compute. OpenAI's o1 (September 2024; OpenAI 2024) and DeepSeek's R1 (January 2025; DeepSeek-AI 2025a) allocate computational resources at inference time to reason before responding, producing substantial capability improvements on reasoning benchmarks.

This paper's hypothesis is that test-time compute serves as a proxy for recursive depth, and that recursive depth may be a fundamental driver of capability amplification in artificial intelligence systems.

Our contributions are:

1. A formal statement of Law I, the ARC Principle, as the ARC Equation $U = I \times R^{\alpha}$, equation (1), with its terms, and definitions of parallel and sequential recursion (Definitions 1 to 3), a contrast drawn earlier, within single models, in the study of test-time compute (Section 2).
2. An estimator of the error-decay exponent α_{err} from two resource levels, equation (3), applied to published reasoning-model data: about 0.1 to 0.3 for parallel sampling and 1.34 from two sequential points. Section 4.2 gives the metric and axis conditions for relating these to capability growth; the comparison remains a hypothesis.
3. The quadratic stability ceiling hypothesis: $\alpha = 2$, the ARC Bound, as the conjectured value of Law III's ceiling $\alpha_{\text{crit}} = 1/(1-\gamma)$, with the domain on which a ceiling of two is equivalent to $\gamma \leq 1/2$ (Proposition 1) and what would refute it (Section 3.3).
4. Falsification criteria for the principle, the three-outcome decision form they inherit, and a programme

hypothesis about human-in-the-loop oversight (Section 5).

Section 2 sets the work against prior work. Section 3 states the principle, its two forms of recursion and the stability ceiling hypothesis. Section 4 gives the analysis of published data, and Section 5 the falsification criteria. Section 6 states what the paper does and does not show, with the programme's status; Section 7 gives its limitations, Section 8 the implications and Section 9 concludes. Appendix A gives the proof of Proposition 1, Appendix B the version history and every correction, Appendix C the priority record, Appendix D the notation and a glossary, and Appendix E the paper's place in the ARC Theory, and Appendix F the dated metric correction and preserved earlier wording.

2 Related work, antecedents and the rival instrument

This section states each debt to prior work in the source's own terms, beginning with the work nearest to this paper's own empirical question, the scaling of test-time compute, then the scaling premise the programme borrows, and ending with the nearest work on how oversight scales and its antecedents.

Test-time compute, studied directly. The scaling of test-time compute in language models has been studied directly, and the contrast this paper draws between sequential and parallel recursion was drawn earlier within single models. Snell et al. (2024), first posted on 6 August 2024, compare sequential revisions with parallel samples under one generation budget and report that "while increased sequential revisions tends to outperform more parallel compute, at higher generation budgets there is an ideal ratio that strikes a balance between the two extremes", the ratio depending on the difficulty of the question. Brown et al. (2024), first posted on 31 July 2024, report that under repeated sampling "the relationship between coverage and the number of samples is often log-linear and can be modelled with an exponentiated power law", while majority voting and reward models "plateau beyond several hundred samples and fail to fully scale with the sample budget". Muennighoff et al. (2025), first posted on 31 January 2025, lengthen one model's thinking by budget forcing, which "allows extrapolating beyond its performance without test-time intervention: from 50% to 57% on AIME24". Zeng et al. (2025), first posted on 17 February 2025, find that the longer chains of thought of o1-like models "do not consistently enhance accuracy" and, comparing "sequential and parallel scaling strategies on QwQ, R1 and LIMO", that "parallel scaling achieves better coverage and scalability". The published evidence within single models is therefore mixed, and the comparison of Zeng et al. (2025), which holds the system fixed, runs against this paper's hypothesis. This paper states the contrast as a hypothesis about the capability exponent in equation (1), reports two-point error-decay estimates in Section 4, and states the stability ceiling of Section 3.3. Relating the error estimates to the capability hypothesis requires the metric and depth bridge of Section 4.2.

The pipes premise, with its contest. The premise that transport networks set the pace of biological growth is not this programme's claim to defend; it is the premise of a leading account in quantitative biology, and a contested one. West, Brown and Enquist derived the allometric scaling laws of biology from the geometry of nutrient-distribution networks (West et al. 1997, 4 April 1997; one of the most cited papers in its field, with an OpenAlex citation count of 5,046 as of 12 August 2026, a single-source and index-dependent count). The exact exponent remains contested: White and Seymour report basal metabolic rate proportional to body mass to the two-thirds power rather than three-quarters (White and Seymour 2003; 725 citations, OpenAlex), and Kozłowski and Konarzewski have twice asked whether the model is "mathematically correct and biologically relevant" (Kozłowski and Konarzewski 2004, 2005). This programme's premise needs only the mechanism, that the delivery network sets the ceiling, and not any particular exponent; the dispute over the exponent's value leaves it untouched, but the critics question

the mechanism too, so the programme takes it as a premise, and its biological reading stands or falls with it.

The exception West's own group found. The escape from the pipe limit is also not this programme's assertion; it is measured in the urban scaling literature by the same senior author. Bettencourt, Lobo, Helbing, Kühnert, and West (2007) (2,704 citations, OpenAlex, 12 August 2026, a single-source count) report, in the paper's own words, read from its abstract (their β is a city-scaling exponent, not this paper's coupling β_L):

> "Quantities reflecting wealth creation and innovation have $\beta \approx 1.2 > 1$ (increasing returns), whereas those accounting for infrastructure display $\beta \approx 0.8 < 1$ (economies of scale)."

Infrastructure, the literal pipes, scales sublinearly. Information-mediated activity scales superlinearly. They state the contrast with biology directly: "we discuss how cities are similar to, and differ from, biological organisms, for which $\beta < 1$". They derive "growth equations, which quantify the dramatic difference between growth fueled by innovation versus that driven by economies of scale" (spelling as in the original). And their closing inference names the consequence: "This difference suggests that, as population grows, major innovation cycles must be generated at a continually accelerating rate to sustain growth and avoid stagnation or collapse."

On this paper's reading, that is the shape of the open problem this programme addresses: in the findings of West's group, information-mediated growth escapes the throughput limit, and the only brake in their framework is external and must be applied again and again, faster and faster. No internal limit is derived. The replacement limit, a ceiling belonging to the growing system itself, is the gap.

The pipes argument at fixed throughput. Everything that grows is fed through a channel, and for everything before software, fixing the channel fixes the growth: starve a tumour's blood supply and its growth is held back; fix a chain reaction's fuel and geometry and it stops; the Eddington limit is a luminosity ceiling where radiation pressure balances gravity, not a hard ceiling on the rate of accretion: ultraluminous X-ray sources sit well above it, one accreting neutron star at about a hundred times it (Bachetti et al. 2014). Cosmic inflation is excluded by scope (expansion of space, not growth of a structure on a substrate); evolution is the sharpest case and fits (complexity rising for billions of years at roughly fixed solar throughput, glacially, with no internal corrector, so on this framework evolution would carry a measurable exponent far below the ceiling, a hypothesis about biology stated without a criterion or a planned test, and outside the theory registration's scope). Software is the first thing that keeps growing when the pipes are held fixed, within the finite number of states and the maximum rate a fixed machine allows (Margolus and Levitin 1998; Lloyd 2000), and the operational definition is load-bearing: an artefact loop on a fixed substrate, measured on a substrate-fixed clock; a model merely thinking longer does not qualify. The physicist's objection is Landauer's (1961), that computation is physically implemented; the fixed-substrate clock is the answer, because the regime under study holds the substrate fixed and asks what still grows.

The rival instrument. The nearest work to this programme's question, and the right paper to weigh it against, is Engels et al. (2025), arXiv:2504.18530, first posted 25 April 2025 at 17:54:27 UTC (arXiv's record; a NeurIPS 2025 Spotlight Poster, checked on 15 August 2026 against the NeurIPS virtual page for poster 115536, as recorded in the statement paper's reference dossier). It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived numerically and in some cases analytically

for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step.

The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, measured through the corrector's own scaling exponent. They also state the design condition in slope form: oversight improves with capability when the overseer's game-specific Elo rises faster with general Elo than the overseen system's ($m_G > m_H$), and they recommend iterating an oversight design until it does. Their framework has no term for what the overseer is made of. The programme's separate prediction, outside this paper's framework, is that iterated same-class oversight, of which Nested Scalable Oversight is an instance, stays capped however many rungs are added, while a cross-class corrector is not held to that cap. The theory registration's P6 states the cap for same-class correctors, at or below one half, as a conjecture on the correction-leverage exponent of Law III (the ARC Ceiling, a law under test), an exponent the registration keeps distinct from Law II's β_C. That is where the two programmes could disagree about a measurable quantity, and this paper does not decide it.

Antecedents and near-misses, each with its differential. *On the question.* Hutter asked directly whether intelligence can explode (Hutter 2012, arXiv, 28 February 2012), undertaking to separate speed from intelligence explosion and to consider possible bounds on intelligence, augmenting the analysis of Chalmers (2010). The question itself goes back to Good (1966). What that literature does not contain, so far as has been surveyed, is a derived ceiling with an architecture dependence; growth exponents of the form used here are older, as the item on the form below concedes, and power laws in inference compute have been fitted directly (the paragraph on test-time compute above). A prior source that supplies a derived ceiling with an architecture dependence defeats the corresponding priority here, and readers who locate one are invited to report it.

On the form. The relation $\alpha = 1/(1 - \beta_L)$ that Figure 1 and Section 5.2 take from the Foundational paper is the exponent of the elementary solution of a separable growth equation. The same structure, a stock that grows at a rate proportional to a power below one of itself, is the knowledge production function of Jones (1995), whose long-run growth rate carries the factor $1/(1 - \phi)$.

On impossibility. Three 2025 arXiv papers argue for limits on alignment or its control: Yao (2025b), "The Alignment Trap: Complexity Barriers" (arXiv:2506.10304, version 1 of 12 June 2025 at 02:30:30 UTC, independently observed by the Internet Archive on 13 June 2025; cited by arXiv:2512.03048); Yao (2025a), "On the Mathematical Impossibility of Safe Universal Approximators" (arXiv:2507.03031, 3 July 2025, the only paper in the arXiv abstract corpus containing the phrase "irreducible uncontrollability", abstract-search total of one, measured 12 August 2026); and Ball et al. (2025), "On the Impossibility of Separating Intelligence from Judgment" (arXiv:2507.07341, 9 July 2025). All three are worst-case and qualitative: measure zero, coNP-completeness, cryptographic hardness. None reports an average-case scaling exponent or a rate. The third concludes that "safety cannot be achieved by designing filters external to the LLM internals (architecture and weights)" and argues that "an aligned AI system's intelligence cannot be separated from its judgment"; that argument from filtering intractability is consistent with this programme's architecture dependence, though it does not test it. Two of the three are by one author. The measured object of all three is verification of a fixed system, and it is closed within its own terms; this programme's measured object is a different one, persistence of alignment as capability compounds under recursive amplification, and closure of the first does not settle the second.

On the mechanism. That anti-correlated estimates average better than independent ones is textbook variance reduction (antithetic variates). The mechanism is not the claim. The claim is that architecture determines whether anti-correlation is available at all, and that this caps a safety-relevant exponent.

The nearest structural analogue. The quantum error-correction threshold theorem (Aharonov and

Ben-Or 2008) also converts a qualitative worry into a critical value. It concerns physical error rates in a fixed architecture, not a corrector's scaling exponent, so it is a near-miss rather than an occupant; the analogy is one of method. This analogue was identified by the programme's own search rather than by a referee, and is disclosed accordingly.

On recursive creation. Smolin's cosmological natural selection (Smolin 1992) is the antecedent for selection-shaped universes, and the programme's own December-era notes cite it contemporaneously ("Echoing Smolin's cosmological natural selection, AI could create recursive universes with their own laws", from the author's notes).

On feedback and stability (classical). The general proposition that inadequate corrective gain relative to system gain causes instability has a long control-theory lineage (small-gain theorems; Zames 1966). The differential: those are gain conditions on interconnected systems, not a power-law exponent criterion on a corrector's scaling with the capability of a recursively improving system. The general proposition is never claimed; the exponent formulation, dynamics and estimator are the contribution surface.

The nearest quantitative neighbour. Liu and Meng (2026), "Self-Correction as Feedback Control: Error Dynamics, Stability Thresholds, and Prompt Interventions in LLMs" (arXiv:2604.22273) recasts self-correction as a closed-loop control problem via a two-state Markov model and derives a directly measurable stability threshold: iterate only when the error-correction rate over the error-introduction rate exceeds $\mathrm{Acc}/(1-\mathrm{Acc})$. The differential: theirs is a per-step rate threshold at a fixed capability level, deciding whether another iteration helps now; this programme's criterion is a scaling relation across capability, deciding whether corrective strength keeps pace as the system improves. Theirs has no scaling exponent on the corrector and no capability-growth exponent, the two quantities the programme's co-scaling criterion compares; the two models meet at zero growth, as Paper X shows. Two adjacent empirical results from the same search: Li (2025), "Decomposing LLM Self-Correction: The Accuracy-Correction Paradox and Error Depth Hypothesis" (arXiv:2601.00828, first posted on 24 December 2025; an accuracy-correction paradox, weaker models showing higher intrinsic correction rates) and Tsui (2025), "Self-Correction Bench: Uncovering and Addressing the Self-Correction Blind Spot in Large Language Models" (arXiv:2507.02778, first posted on 3 July 2025; a systematic self-correction blind spot across open models).

3 The ARC Principle and its framework

3.1 The ARC Equation

The ARC Principle (Artificial Recursive Creation), stated in an earlier form, as the ARC Hypothesis, in the author's manuscripts of 8 December 2024 (Eastwood 2024) and articulated in *Infinite Architects* (Eastwood 2026a), where its exponent is two, states, in the form this paper gives it:

$$U = I \times R^{\alpha} \tag{1}$$

Capability scales with intelligence multiplied by recursive depth raised to a power, where:

- U = Effective capability on a specified metric (benchmark performance requires the metric bridge of Section 4.2; the letter's earlier senses in this programme are in Appendix D.2)
- I = Base intelligence (single-pass processing capacity without recursive reasoning)
- R = Recursive depth (number of self-referential processing iterations)
- α = Scaling exponent (empirically determined; hypothesised stability ceiling = 2, a scaling limit above which self-correction is not stably maintained)

The principle's core claim: recursion does not merely add to capability; it multiplies it according to a power law. The letter α has an earlier sense in this programme, recorded in Appendix D.2.

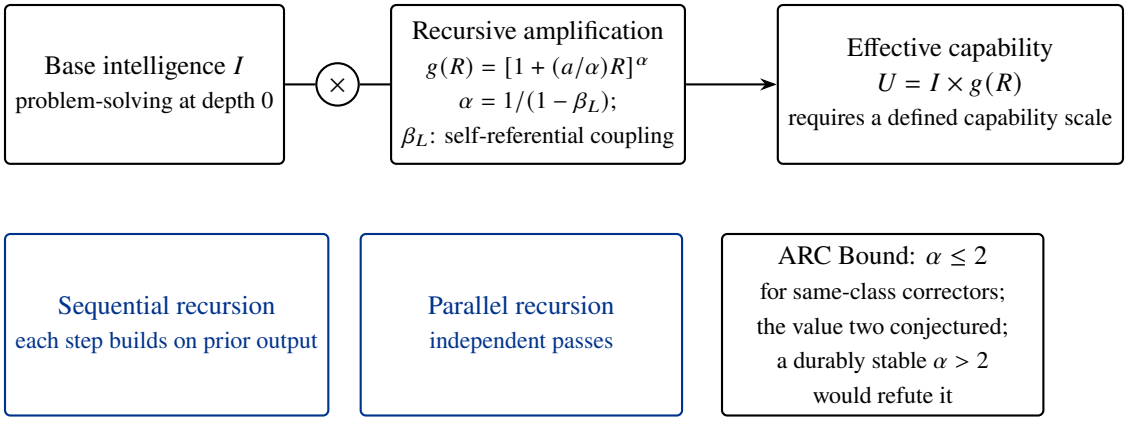


Figure 1. The architecture of the ARC Principle. $U = I \times g(R)$ with $\alpha = 1/(1 - \beta_L)$, where β_L is the growth model's self-referential coupling, not Law II's correction exponent β_C. In $g(R) = [1 + (a/\alpha)R]^{\alpha}$, $a > 0$ is a rate constant and R counts passes after the first: g solves $dg/dR = ag^{\beta_L}$ with $g = 1$ before an added pass, the equation of the Foundational paper, which writes that count $r = R - 1$. Equation (1) counts the first pass as $R = 1$. For $\beta_L < 1$, the displayed α is the shifted solution's asymptotic exponent; a finite-window fitted slope need not equal it. The recursion architectures and the conjectured ARC Bound are illustrated, not located by an empirical coupling measurement. For orientation, Paper II's Gemini 3 Flash sequential regression gives an error-decay exponent of 0.49 with standard error 0.20, one model's inconclusive fit; this paper's earlier sequential endpoint estimate is ∼1.34, and its parallel error slopes are about 0.1 to 0.3 (Section 4). Section 4.2 states the additional metric and axis premises needed to read these as capability slopes. Cauchy Unification (Paper VII, under its stated axioms) uses $\alpha = d/(d + 1)$ for physical networks of dimension d, a formula earlier work derived (among them West et al. 1997), and the growth-model relation $\alpha = 1/(1 - \beta_L)$ for intelligence. Those model relations do not convert an error fit into an independently measured coupling. The ARC Bound is at most two for correctors of the system's own class, as registered, with the point value two conjectured. The earlier $\alpha \approx 2.24$ is retracted: it came from the DeepSeek R1 run of 21 January 2026, the first phase of Paper II (Paper IX, Section 7). Thirteen falsification criteria stand across the Foundational paper; this paper's four original criteria are in Table 4. Sources: this paper and Papers II, III, VII and IX.

3.2 Recursion and its two forms

Definition 1 (Recursion)**.** Recursion is self-reference: a process whose output becomes its input. It is distinct from mere iteration (repeating the same operation) because each cycle operates on the transformed results of previous cycles.

Definition 2 (Parallel recursion, weak)**.** Multiple independent solutions generated simultaneously. No information transfer between branches. Example: generating N samples and selecting by majority vote. Expected scaling: diminishing returns as redundancy increases.

Definition 3 (Sequential recursion, strong)**.** Each processing step builds explicitly on previous steps. Errors can be detected and corrected iteratively. Example: chain-of-thought reasoning (Wei et al. 2022) that folds self-reflection into each step. Expected scaling: compounding returns as depth enables self-correction.

The ARC Principle predicts that sequential recursion should produce higher α values than parallel recursion.

3.3 The quadratic stability ceiling hypothesis

We hypothesise that $\alpha = 2$ represents the stable limit, the most a recursively self-correcting system can grow and stay correctable: a scaling limit, not a speed limit. The exactly quadratic speedup belongs to the quantum search algorithm of Grover (1996); Bennett et al. (1997) then established the lower bound that matches it, which is what makes that speedup optimal for unstructured search. The correspondence invoked here is structural, the same quadratic form of a search-optimality bound, not a claim that recursive reasoning implements amplitude amplification. If a similarly shaped limit applies to recursive intelligence for related reasons, quadratic scaling may mark the ceiling above which self-correction is not stably maintained; systems can be driven beyond it, they do not remain stably self-correcting there, and any excursion above 2 is the predicted supercritical regime rather than a refutation of the bound. A durably stable system above 2 would refute the bound. Stable is measured, not asserted: correction has to out-scale drift, which Law II states as $\beta_C > k$, and the lower bound on $\beta_C - k$ has to sit above zero throughout a window fixed before the run began.

Law before value. The ceiling of stable self-improvement is set by how fast a system can correct itself: it is the reciprocal of one minus the correction-leverage exponent γ,

$$\alpha_{\text{crit}} = \frac{1}{1-\gamma}. \tag{2}$$

For a corrector built from the same substrate as the engine, that reciprocal is two at $\gamma = 1/2$, the ARC Bound's point value, which the theory registration conjectures; what it claims is the inequality, a ceiling of at most two for correctors of the system's own class. The point value rests on two assumptions the registration names and does not present as established: the burden model, under which the drift exponent follows the rate of capability gain, and the route to one half, that internally accumulated corrections combine like independent samples. The law is the claim. Two is what the law returns at $\gamma = 1/2$, and the conditions behind one half are what the drafted and registered units examine: Unit 82, registered on 27 October 2026, tests P17, which bears on one of them and establishes neither the value nor the bound (Section 6.3). If two dies, the law survives completely: measure γ and report the ceiling it implies. If the law dies, two has no independent support, because the derivation runs through $1/(1-\gamma)$. Version 1.104 of the theory registration states the relation as "a conditional specialisation of the second law, not a third finding standing beside it", on which a result is "never independent corroboration of the second law". The relation is derived inside a pacing model and has not been derived from a joint model of capability growth, correction accumulation, error covariance, correction delay and absolute harm, so Law III is stated here as a minimal-model result rather than as a general law. One consequence should be met where the law is met, because it cuts both ways: since, on the domain $\gamma < 1$ of Proposition 1, $1/(1-\gamma) \leq 2$ is *exactly* equivalent to $\gamma \leq 1/2$, the ceiling and the half-value are one commitment written in two notations rather than two independent claims, and neither may ever be offered as evidence for the other.

Proposition 1 (The domain of the equivalence)**.** *For $\gamma < 1$, $1/(1-\gamma) \leq 2$ holds if and only if $\gamma \leq 1/2$. At $\gamma = 1$ the reciprocal inequality has no truth value while $\gamma \leq 1/2$ is false, and for $\gamma > 1$ the reciprocal inequality holds while $\gamma \leq 1/2$ fails. The equivalence therefore holds exactly on $\gamma < 1$, the range on which $\alpha_{crit} = 1/(1-\gamma)$ is positive and finite.*

The proof is in Appendix A. The restriction is the model's own rather than an assumption added to it, and it is what the clauses above already imply where they state that the relation is derived inside a pacing model and is offered as a minimal-model result rather than as a general law. Read inside that domain, the point of the sentence stands entire.

4 Empirical analysis

4.1 Data sources

The parallel series. OpenAI's account of o1, *Learning to reason with LLMs* (12 September 2024; OpenAI 2024). Benchmark: AIME 2024 (American Invitational Mathematics Examination). Variable: number of samples: a single sample, consensus among 64 samples, and re-ranking of 1,000 samples with a learned scoring function.

The sequential series. DeepSeek-R1 and its update DeepSeek-R1-0528, as the developer's model card for DeepSeek-R1-0528 reports them (28 May 2025; DeepSeek-AI 2025b); the model is described in its technical report, arXiv:2501.12948 (DeepSeek-AI 2025a). Benchmark: AIME 2025. Variable: average thinking tokens a question (chain-of-thought length). The two points are two versions of the model, each at its own average token use, not two thinking budgets of one version.

4.2 Method

Equation (3) estimates an error-decay exponent, denoted α_{err} here, over a declared resource coordinate R_{res}. For positive errors at two distinct positive resource levels, it is the slope implied by $E(R_{\text{res}}) \propto R_{\text{res}}^{-\alpha_{\text{err}}}$. In the parallel series the resource is sample count; in the sequential pair it is the developer's average thinking-token count. Neither coordinate is a measured count of completed self-improvement revisions. This is the same endpoint estimator used in Paper II, version 2.19, equation (2); its equation (1) models the proportion of incorrect responses.

$$\alpha_{\text{err}} = -\frac{\ln(\text{Error}_2/\text{Error}_1)}{\ln(R_{\text{res},2}/R_{\text{res},1})} \tag{3}$$

The metric bridge. On the same resource coordinate, defining a proxy as $C_{\text{err}} = 1/E$, or a fixed positive multiple of it, gives the same numerical exponent when the stated error power model holds and $E > 0$. Accuracy, $1 - E$, is a different response and approaches its ceiling; it does not thereby grow as a power with that exponent. Identifying α_{err} with the capability exponent α in equation (1) therefore requires an explicitly justified capability metric and a mapping from the resource coordinate to recursive depth. A change of response scale or input axis can change the exponent. This distinction follows the Operational Definitions register, version 2.0.0 (1 November 2026), entries EQ-01 and EQ-07. The power-law slope alone identifies no self-improvement or coupling mechanism.

4.3 Parallel recursion

Table 1. Parallel recursion: accuracy and error rate of o1 on AIME 2024 by number of samples (a single sample, consensus among 64 samples, re-ranking of 1,000 samples), as OpenAI reports them (OpenAI 2024).

Samples (R_{res})	Accuracy (%)	Error Rate (%)
1	74	26
64	83	17
1000	93	7

Finding. The parallel series gives error-decay slopes $\alpha_{\text{err}} \approx 0.1$ to 0.3, sub-linear for the reciprocal-error proxy on this sample axis under the power model. This does not identify a capability-growth exponent without the bridge above.

4.4 Sequential recursion

Table 2. Sequential recursion: accuracy and error rate on AIME 2025 by average thinking tokens a question, for two versions of DeepSeek-R1, the second row being DeepSeek-R1-0528, as the developer reports them (DeepSeek-AI 2025b).

Thinking Tokens (R_{res})	Accuracy (%)	Error Rate (%)
~12,000	70	30
~23,000	87.5	12.5

Finding. The sequential pair gives an endpoint error-decay slope $\alpha_{\text{err}} \approx 1.34$. Under the stated reciprocal-error proxy and power model, that is a super-linear proxy exponent over the token axis; it is not a direct measurement of capability growth or coupling. The average token count differs by roughly a factor of 1.9, below the span of at least one and a half decades of recursive depth, a factor of about thirty, that version 1.104 of the theory registration requires of an admissible exponent measurement. That registered requirement concerns the declared depth instrument; a token span alone would not establish the necessary axis mapping. The two points are also two versions of the model (Section 4.1), so the reading that each layer of reasoning amplifies previous gains remains a hypothesis. The later cross-architecture measurement, on five frontier models, reports architecture-dependent behaviour (Paper II, version 2.19). On Gemini 3 Flash, its one interpretable fit, it reports an endpoint sequential error-decay exponent of 0.59 with a bootstrap 95 per cent interval of -1.3 to 2.9, a regression sequential error-decay exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a parallel error-decay exponent of 0.31 by endpoint estimate, the two arms not matched in total compute. The sequential regression uses three non-zero-error levels of one fixed-weight model; reasoning settings and prompt wording also vary. These are one model's test-time resource fits and are inconclusive. They identify no independent capability-growth or coupling measurement. The earlier 1.34 is retained as the paper's original fitted value, not a programme-wide capability exponent.

4.5 Summary of findings

Table 3. Error-decay slopes by method and resource proxy, using equation (3). Sub-linear and super-linear classify the reciprocal-error proxy under the same power model, not an independently identified capability scale.

Method	Recursion Type	Fitted α_{err}	Classification
o1 (1 to 64)	Parallel	0.10	Sub-linear
o1 (64 to 1000)	Parallel/Hybrid	0.32	Sub-linear
DeepSeek-R1	Sequential	~1.34	Super-linear reciprocal-error proxy (two points; later error fits in Paper II)

Key finding, held as a hypothesis. The form of recursion may affect an error-decay slope on a fixed instrument. The present comparison does not isolate that effect or identify a common capability scale.

What the comparison holds fixed. The two rows this comparison rests on differ in more than the form of recursion. The parallel row comes from a system whose resource variable is the number of samples, under consensus voting and then re-ranking, reported in September 2024 (OpenAI 2024); the sequential row comes from a different system, built by a different developer, whose resource variable is an average thinking-token count, reported on 28 May 2025 for two versions of the model (DeepSeek-AI 2025b). The benchmarks differ too: the parallel series is on AIME 2024 and the sequential pair on AIME 2025. The system under test, the benchmark, the evaluation context and the quantity on the horizontal axis vary together across the pair. A difference in fitted exponent between them is consistent with a role for the form of recursion, and it is equally consistent with those other differences, so the key finding above is to be read as a hypothesis rather than as an inference from the two rows. The comparison that would isolate the form of recursion holds the system and the resource definition fixed and varies only the form, and Section 5 is where its outcome is recorded, at F1 and F4. Comparisons within single models have been published (Section 2): Snell et al. (2024) report that sequential revisions tend to outperform parallel samples, with a balance of the two best at higher budgets, and Zeng et al. (2025) find parallel scaling ahead on three o1-like models, R1 among them, so the published evidence is mixed. Both tables, both fitted values and the direction of the difference are unchanged.

5 Falsification

5.1 Criteria and their status

The following criteria concern the original capability hypothesis. Applying an error-decay estimate to them first requires the metric and depth bridge of Section 4.2; none is a criterion on an arbitrary error scale. Subject to that requirement and the registered decision form below, the ARC Principle (Law I, like every law of the theory a law under test) would be significantly weakened or refuted if:

Table 4. Falsification criteria of the ARC Principle in this paper, with their current status.

Code	Condition	Current Status
F1	Sequential recursive depth consistently yields $\alpha \leq 1$	Undecided: Paper II, version 2.19, reports an error-decay regression of 0.49 (standard error 0.20) and endpoint estimate 0.59 (bootstrap 95 per cent interval −1.3 to 2.9) on Gemini 3 Flash. This one-model fit is inconclusive and the capability/depth bridge is not independently established. Estimates above 1.0 from ceiling-compressed data may be inflated. Neither direction of "consistently" is decided
F2	α decreases as recursive architectures mature	Undecided: Paper II's cross-architecture study leaves only Gemini 3 Flash in measurable error range; more capable models reached the ceiling and a less capable one the floor. It supplies no identified trend in the capability exponent across architectures
F3	The relationship is additive rather than multiplicative	Not met; no result yet bears on it
F4	More extensive datasets show $\alpha < 1$ for sequential reasoning	Undecided: the two Gemini 3 Flash error-decay point estimates in Paper II are below one, but the endpoint interval spans both regimes and the capability/depth bridge is unestablished. This does not partially decide a criterion on capability growth. The artefact-mediated estimation written for this paper (unit 24) is a draft preregistered by timestamp (first anchored on 12 August 2026 at block 962066), awaiting submission by the author

5.2 What would refute the principle

The ARC Principle in this paper's original, stronger form, that capability scales super-linearly with recursive depth, $U = I \times R^{\alpha}$, with the sequential/parallel distinction determining the regime, is refutable. It would be overturned by any of the following:

1. **No super-linear depth scaling.** If capability scaled only linearly or sub-linearly with recursive depth R across reasoning models (i.e. $\alpha \leq 1$ where the principle predicts $\alpha > 1$ for sequential recursion), the central super-linearity claim fails.
2. **Exponent-composition mismatch.** Given independently measured coupling β_L, a declared capability metric and depth axis, and the growth-model premises and asymptotic regime of Figure 1, systematic disagreement with $\alpha = 1/(1-\beta_L)$ would reject that model's applicability to the measured system. It would not refute the algebraic solution under its premises. Inferring β_L from the same fitted exponent and substituting it back supplies no such test. The coupling remains distinct from Law II's correction exponent β_C.
3. **Regime non-distinction.** If sequential and parallel recursion produced the *same* scaling regime (no super-linear vs sub-linear split), the paper's central distinction collapses.
4. **Better-fitting alternative.** If a functional form outside the ARC family fit the test-time compute data materially better across models, ARC would not be the operative law.
5. **Domain-independence failure.** If $U = I \times R^{\alpha}$ held for reasoning models but robustly failed in other recursive domains meeting the premise, the domain-independent claim would have to narrow to *"a property of current reasoning models"*.
6. **The one-half reference.** On Gemini 3 Flash, Paper II's error-decay regression is 0.49 (standard error 0.20), one model's inconclusive fit (Section 4.4). One half is a reference value, not an automatic null exponent for the proportion of incorrect responses: the central limit theorem's square-root standard-error scaling does not by itself supply a law for that proportion. A competing sampling, aggregation and decision model must predict the same measured outcome before a comparison can distinguish mechanisms. The earlier objection is retained with its correction in Appendix F; its programme record is FALS-022 and Objection 5 of Paper II, Section 5.4.

This is the metric correction recorded by Paper II on 19 September 2026.

5.3 The decision form

The decision form these criteria inherit. Section 5.1 records status against this paper's original criteria; the deciding registrations use a three-outcome form, and every criterion above inherits it: SUPPORTED when the interval lies wholly beyond the minimum effect fixed in advance; REFUTED when the interval lies wholly within the equivalence margin fixed in advance (two-one-sided-tests logic); INSUFFICIENT PRECISION when the interval is wider than the margin, reported in exactly those words and never as support or refutation. The margin, the minimum effect and a measured power figure are fixed before filing. "Refuted if the confidence interval contains the null" lets an underpowered study refute by default and is not licensed anywhere in this programme.

5.4 An objection pre-empted

One objection pre-empted, because a physicist reaches it in one step. Fault-tolerant quantum error correction suppresses logical error exponentially in code distance below threshold, which looks like a counterexample to any square-root cap (the same-class limit $\gamma \leq 1/2$ of Section 3.3). Two scope facts answer it: the exponential suppression runs on the redundancy axis at fixed capability, not on the coupling

of corrective strength to the capability of the corrected system, which is the exponent this programme measures; and the threshold theorem itself requires sufficiently uncorrelated physical noise, which is the independence premise again, wearing hardware. Concentration bounds likewise give exponentially falling error probability while estimator width falls as the square root: different quantities, and conflating them manufactures a refutation.

5.5 A programme hypothesis beyond this paper

Oversight arrangements that place a human in the loop, as amplification (Christiano et al. 2018) and reinforcement learning from human feedback (Christiano et al. 2017) do, are cross-class by construction, because the human corrector does not share the model's substrate. The programme therefore hypothesises, beyond what this paper's framework decides, that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems. Both the operational definition of a correction exponent under oversight and the rules that settle the comparison come from the corrector-class units, namely the corrector-class audit (unit 06) and the corrector-class contrast (unit 01). The audit was registered on OSF on 1 October 2026 and has been public since 4 October 2026; Section 6.3 gives its run. The contrast is a draft, preregistered by timestamp (first anchored on 28 August 2026 at block 964465), awaiting submission by the author.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

6 What this paper does and does not show

6.1 The claims and their status

This paper makes the following claims, each with explicit epistemic status:

Table 5. This paper's claims, each with its status and its level of evidence.

Claim	Status	Evidence Level
$U = I \times R^{\alpha}$ is a useful framework for AI systems	UNDER TEST (Law I)	Theoretical; registered as Law I on 8 September 2026 (the theory registration)
Parallel sampling gives $\alpha_{\text{err}} < 1$ in the cited series	PRELIMINARY	Error decay against sample count (o1); a reciprocal-error proxy has this exponent under the power model. It is not an independently identified capability exponent
Sequential recursion yields capability exponent $\alpha > 1$	UNRESOLVED	The original 1.34 is an error-decay endpoint estimate from two versions of DeepSeek-R1. Paper II's later error-decay regression 0.49 (standard error 0.20) and endpoint estimate 0.59 (bootstrap 95 per cent interval −1.3 to 2.9) concern one interpretable model, Gemini 3 Flash. They are inconclusive and require the capability/depth bridge (Section 4.2)
$\alpha = 2$ is a stability ceiling: exceedable, but not while a system remains stably self-correcting	HYPOTHESISED	Theoretical only

Claim	Status	Evidence Level
The form of recursion matters	HYPOTHESISED	Consistent with both datasets, which also differ in system, benchmark, evaluation context and resource axis (Section 4.5)

This paper does not claim:

- That $U = I \times R^2$ applies to cosmological or universal scales (that application remains speculative; on the limits physics places on computation, see Lloyd 2002)
- That a capability exponent of 1.3 is established by the earlier error fit (the metric and depth bridge, as well as further data, is required)
- That the principle has been independently replicated (it has not)

We present a principle with preliminary and mixed evidence (Sections 4 and 5), and invite rigorous testing.

6.2 The exponent-composition identity

The programme's stability ceiling and its conversion exponent meet at the same number: the ceiling $\alpha_{\text{crit}} = 1/(1-\gamma)$ of equation (2) and the exponent $\alpha = 1/(1-\beta_L)$ of Figure 1 share one functional form and agree exactly where $\beta_L = \gamma$ on the same system, and the temptation is to say that one constant does two jobs. The claim's status is stricter than that and must be stated strictly: the identity is PREREGISTERED BY TIMESTAMP FOR TESTING, in advance of the data, with outcomes that can kill it. Two drafted registrations were written to carry it. One of them, Study AE, was withdrawn on 12 August 2026, before filing, as vacuous: a shared accumulation law forces the agreement it would have tested, so its test could not fail, and the programme's register names the cross-class to same-class correction-exponent ratio as its replacement carrier. Study AE set out the reciprocity requirement between the separately measured leverage cap and stability boundary, on the log scale, with three outcomes per cell (supported, refuted, insufficient precision) and an aggregation rule under which one refuted cell refutes the identity; its independence structure was disclosed in advance, including enumeration of any endpoint overlap between the two feeder designs, so that common-method variance would be assessed before analysis rather than raised afterwards as the explanation for agreement. Study AG, a draft preregistered by timestamp (first anchored on 12 August 2026 at block 962066), estimates the coupling and the correction exponent on one and the same system and tests whether the growth exponent the leverage fraction sustains respects the ceiling the correction exponent sets ($\beta_L \leq \gamma$), and is designed to find a violation.

So the identity is stated here as a claim preregistered by timestamp for testing, its deciding measurements drafted and dated; if the channels disagree, the unification is withdrawn from every surface it has reached, with the same prominence as the framing. The identity is never called established. Paper II's Gemini 3 Flash regression value 0.49 (standard error 0.20; Section 4.4) is an error-decay slope, not a measurement of β_L or γ. Comparing it directly with an asserted one half would compare differently defined quantities. Even after choosing a reciprocal-error capability proxy, identifying its finite-window slope with the asymptotic exponent in $\alpha = 1/(1-\beta_L)$ requires the growth-model and axis premises of Figure 1. Algebraically inverting an assumed identity reuses the fitted slope and supplies no independent coupling measurement or corroboration; a comparison with the correction-leverage exponent needs its own instrument on the same system.

6.3 Laws IV and V, and the programme's status

This paper tests neither Law IV nor Law V. No result in this paper is a deciding test of P15, P12 or P21; the note of 3 October 2026 that set out why is kept in Appendix B.3.

Status, as at 1 November 2026. What this programme names Laws are conjectures. Laws I to III were registered on 8 September 2026 and Laws IV and V on 15 October 2026 (version 1.103), the mathematics of Laws IV and V still in development (version 1.2, 29 October 2026). Each is a law under test: Law IV has not yet been tested, and the counted test of Law V's placement limb is Unit 83, below. The theory registration of twenty-two propositions was registered on 8 September 2026 at version 1.100 and updated on 13 September 2026 (version 1.102), 15 October 2026 (version 1.103) and 29 October 2026 (version 1.104, which adds no prediction and changes no law's statement); its terms are defined in a separate register. The first registered study, Unit 06, a classification audit that tests no law, was registered on 1 October 2026 and ran on 6 October 2026; one of its two runs could not start, so it is reported in full as administratively incomplete, with no inferential verdict. A second, Unit 83, the counted test of P21 under Law V, was registered on OSF on 22 October 2026 and ran on 27 and 28 October 2026: P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment (results: DOI 10.17605/OSF.IO/98HKR). Two different axes are named here. The access axis is what Unit 83 varied: both reviewers are called each round, but the corrector sees the earlier kept entries with a marked new arrival and may revise it, while the monitor sees its accumulated register with the new arrival unmarked and can only accept or reject it. The performance axis is what it measured, the outcome, as against the mechanism axis, which was not evaluable. Law V, the ARC Embedding Law, remains a law under test: its placement limb, P21, is supported on the performance axis only, with its mechanism axis not evaluable, which is not support for the information-asymmetry mechanism; its build-order limb, read on P12, is undecided, since P12 is untested. A third, Unit 82, the counted test of P17, the premise beneath Law III's value two (Section 3.3), was registered on OSF on 27 October 2026 and has not yet run. The trials that would decide this paper's own predictions, the artefact-mediated estimation written for it among them (row F4 of Table 4), are drafts awaiting the author's submission. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

7 Limitations

Scientific integrity requires explicit acknowledgement of limitations:

- **Limited data points:** The DeepSeek-R1 analysis relies on only two data points, from two versions of the model (Section 4.1). Additional measurements would strengthen or refute the finding.
- **Token counts as averages:** The thinking-token counts are the developer's published averages a question, about 12,000 for the earlier version at 70% and about 23,000 for DeepSeek-R1-0528 at 87.5%; no count for each problem is published.
- **Domain specificity:** Current evidence is limited to mathematical reasoning (AIME 2024 and AIME 2025). Generalisation to other domains remains untested.
- **Confounded comparison:** The parallel and sequential series differ in system, developer, benchmark, evaluation context and resource axis as well as in the form of recursion (Section 4.5), and published comparisons within single models are mixed (Section 2); the contrast is a hypothesis.
- **Functional form:** Two points fix the exponent of a power law through them and cannot tell a power

law from a logarithmic, exponential or saturating form; equation (3) assumes the power law, and the registered analysis of Law I compares it with rival forms fixed in advance.

- **Metric and axis:** Equation (3) measures error decay against samples or average tokens. Reciprocal error has the same exponent only on the same axis under the stated positive-error power model; accuracy does not. A capability interpretation requires a justified metric and mapping to recursive depth, and a finite-window exponent need not equal the growth model's asymptotic exponent. Neither algebraic inversion nor fixed model weights independently measures coupling.
- **Developer-reported figures:** Every number in Tables 1 and 2 is the developer's own published figure; none was measured for this paper.

8 Implications

8.1 For AI development

If the ARC Principle holds, recursive depth constitutes a third scaling axis alongside parameters and data. Investment in recursive architectures may yield better returns than scaling model size alone.

8.2 For AI safety

If recursion amplifies not only capability but also embedded values, then well-aligned initial values should strengthen through recursive self-improvement. Misaligned values would also compound, making early alignment critical. The paper measures nothing about values and nothing about usefulness (legitimate task completion, needless refusal, resource use, correction cost), so neither half of that conditional says whether a system whose values compound stays useful. Neither half is a reason to relax oversight, containment or shutdown, which stay in human hands for as long as they can be held; correction built in is the contingency for control lost against our will.

8.3 For scientific understanding

The ARC Principle connects to several frameworks, including the scaling laws of Kaplan et al. (2020), Integrated Information Theory (Tononi 2008) and the optimality proof for Grover's quantum search (Bennett et al. 1997).

9 Conclusion

We have formalised the ARC Principle and presented preliminary evidence:

1. Parallel sampling gives error-decay slopes $\alpha_{\text{err}} \approx 0.1$ to 0.3 on the sample-count axis
2. The sequential pair gives $\alpha_{\text{err}} \approx 1.34$ on the average-token axis, from two versions of one model. Paper II's later one-model interpretable error fit is inconclusive. Neither independently identifies a capability exponent or coupling; the reciprocal-error reading needs the metric, power-model and axis premises in Section 4.2
3. The form of recursion may determine whether capability compounds, which the paper holds as a hypothesis (Section 4.5)

In plain terms, the original stronger hypothesis claims: "Thinking about thinking makes you smarter. Not linearly smarter, but disproportionately smarter, if the thinking is sequential rather than parallel."

The principle stands as a law under test. The research continues.

Code and data. The code and the recorded outputs behind this paper are in the folder `experiments` of the public repository github.com/MichaelDariusEastwood/arc-principle-validation. All contributions welcome, including falsifications. The paper and its earlier versions are deposited at doi.org/10.17605/OSF.IO/B6N27.

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Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

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A Proofs

The proof of Proposition 1, in the words of the note of 19 September 2026 that first printed the domain (kept whole in Appendix B.3).

Proof of Proposition 1. For $\gamma < 1$ the quantity $1-\gamma$ is positive, so $1/(1-\gamma) \leq 2$ multiplies to $1 \leq 2(1-\gamma)$, that is $\gamma \leq 1/2$, and the implication reverses by the same step: the equivalence holds in both directions. At $\gamma = 1$ the quantity $1 - \gamma$ is zero, so $1/(1 - \gamma)$ is undefined and the reciprocal inequality has no truth value, while $\gamma \leq 1/2$ reads $1 \leq 1/2$ and is defined and false. Exactly one side is undefined, and the displayed equivalence is therefore not defined at that point. For $\gamma > 1$ the quantity $1 - \gamma$ is negative, so $1/(1 - \gamma) < 0 < 2$ holds for every such γ while $\gamma \leq 1/2$ holds for none, and the equivalence fails. The domain $\gamma < 1$ is exactly the range on which $\alpha_{\text{crit}} = 1/(1 - \gamma)$ is positive and finite and can serve as a ceiling at all. □

B Version history and corrections

B.1 Version 1.17

Version 1.17, 31 October 2026. The paper is set in the programme's common form for its papers: a title block with its category and its place in the programme, an abstract within arXiv's limit, numbered sections with the related work, the claims, the limitations and the conclusion in their own sections, three numbered definitions and a proposition with its proof in Appendix A, captioned tables, a redrawn figure, and lettered appendices for this history, the priority record, the notation and the paper's place in the ARC Theory. The headline question is now the title; the earlier titles are in Appendix B.4. Substantive: (i) Law I is stated as version 1.104 of the theory registration states it, a power law in recursive depth with the exponent measured rather than assumed, and this paper's super-linearity claim as its own original, stronger form; (ii) the two published series are given their sources as their developers published them: the parallel series from OpenAI's account of 12 September 2024, not the o1 system card, which does not contain it, and the sequential pair from the model card for DeepSeek-R1-0528 of 28 May 2025, on AIME 2025, two versions of the model each at its own average token use, where version 1.16 gave the DeepSeek-R1 report and AIME 2024 for both; the token counts are the developer's averages, so "estimated" leaves Table 2 and the second limitation; no number changes, and the comparison of the two rows, already held as a hypothesis, gains one more reason; (iii) the claim that the form of recursion matters is held as a hypothesis in Table 5, the abstract and the conclusion, where version 1.16 gave it SUPPORTED, a status the paper's own decision form excludes, and the first claim's status reads UNDER TEST (Law I), where version 1.16 printed a label that understated a law registered on 8 September 2026; (iv) the stability ceiling: the value two is what Law III returns at $\gamma = 1/2$, a point value the registration conjectures and that rests on the two assumptions it names, where version 1.16 said the law predicts two under one assumption; the sentence calling stability Law II's gives way to the paper's own statement of what stable means, moved from its abstract; and the domain of the equivalence is stated as Proposition 1; (v) the status: Laws IV and V were registered as laws on 15 October 2026 (version 1.103), the registration's current version is 1.104 (29 October 2026), Unit 83's registered result on P21 and Unit 82's registration are given (Section 6.3), Unit 06 is stated once, and the mathematics of Laws IV and V is cited at version 1.2; (vi) in Section 2 each work is stated in its own words: Ball and co-authors' conclusion as their abstract gives it, where version 1.16 quoted words their abstract does not contain; Bettencourt and co-authors' symbols as printed; Kozłowski and Konarzewski's question in their own title, and the mechanism as one the same critics contest; Engels and co-authors' slope condition; Liu and Meng where the two models meet; Chalmers, Smolin, the threshold theorem and the small-gain theorems referenced; the Eddington limit as a luminosity ceiling, not a ceiling

on accretion; software's growth at fixed pipes within the bounds a fixed machine allows; and the evolution exponent as a hypothesis outside the registration's scope; (vii) the prediction about human-in-the-loop oversight is stated as a programme hypothesis beyond this paper's framework; (viii) the principle is dated from the author's manuscripts of 8 December 2024, which state it in an earlier form as the ARC Hypothesis, and the book is named as where it was articulated, with the exponent two; (ix) Paper II's later study is described as a study on five frontier models in its capability arm, and its parallel exponent as an endpoint estimate, not compute-matched; (x) the zero-parameter null carries Paper II's review of 19 September 2026, and its pointer follows Paper II's numbering; (xi) Section 2 opens with the work on test-time compute, in which sequential and parallel compute were compared within single models before this paper (Snell and co-authors; Zeng and co-authors), and records that the published evidence is mixed, one comparison running against this paper's hypothesis; contribution 1, Section 4.5 and the limitations say so, and the related work no longer says that its literature holds no measurable exponent: the question goes back to Good, growth exponents of this form to Jones's knowledge production function, and power laws in inference compute have been fitted; (xii) Section 6.2 gives study AG's test as its registration text now states it, whether the coupling respects the ceiling the correction exponent sets, where version 1.16 said that the two derivations agree only where both parameters equal one half and disagree in their derivatives everywhere else, which is not so, since the two share one functional form; (xiii) Section 3.3 states Law III's standing in the words of version 1.104 of the registration, a conditional specialisation of Law II that is never independent corroboration of it, where version 1.16 said that the form is settled and its depth is not, and the equivalence of a ceiling of two with $\gamma \leq 1/2$ is stated on its domain where it is first stated; (xiv) the conclusion's last line, the key finding of Section 4.5 and the closing sentence of Section 6.1 carry their status, the principle as a law under test, the finding as a hypothesis and the evidence as preliminary and mixed, where version 1.16 printed "The principle stands", "Key finding" and "preliminary supporting evidence"; (xv) Section 7 adds the confounded comparison, the problem of functional form at two points and the provenance of every number; rows F2 and F3 of Table 4 say what bears on them, where both read "Not met"; Section 4.2 states the power law that equation (3) assumes; Figure 1 defines its rate constant and its count of depth, and states the ARC Bound as registered and the formula $\alpha = d/(d+1)$ as earlier work derived it; and Section 4.4 gives the span of depth that version 1.104 of the registration requires of an exponent measurement, one and a half decades, where version 1.16 gave one decade. Editorial: dated notes the body kept beside text corrected in this version move whole to Appendix B.3; the correction note of 3 October 2026 stands beneath the 1.14 entry, whose pointer to it is repointed (its last words, the only words of a dated entry's text that change); the record of dated predictions moves to Appendix C.2; Paper II's row for Gemini 3 Flash is given whole once (Section 4.4) and briefly elsewhere; every mention of one of the author's papers links its OSF DOI, and every cited work its official source; internal labels of the programme's source checks are not set; the author's notes are named as his; in Section 2 the pipes premise is given as a contested premise and its inference as this paper's reading; the optimality proof for Grover's search is credited to Bennett and co-authors (Section 8); Section 4.4 holds the reading of the two sequential points as a hypothesis and gives Paper II's finding in its own words; Figure 1 gives the parallel regime by this paper's own series and the retracted estimate of about 2.24 by its run, and does not carry the artwork's figures this paper does not report; the programme's stance that oversight, containment and shutdown stay in human hands, and the paper's silence on usefulness, are stated in Sections 1 and 8.2; two sentences now state the author's hypothesis and the stable limit as his own (Sections 1 and 3.3); the plain-terms line of Section 9 is given as the principle's claim; the notes on the earlier senses of two letters move to Appendix D.2; the authorship sentence moves from the Acknowledgements, which hold the author's disclosure alone, to the copyright block, which takes the programme's common form with the assertion under the Copyright, Designs and Patents Act 1988; the clause that followed that assertion, citing guidance of the United States Copyright Office, and the link to a fuller statement are not set (version 1.16 prints them); Section 6.2 defines the identity it discusses and no longer sets the programme's publishing

rules in the paper's voice; Appendix D.2 no longer points to website pages; "sealed" in Appendix C.2 carries a bracketed note; Wei and co-authors are cited for chain-of-thought reasoning alone; the three impossibility papers of Section 2 carry their measured object, and the sentence on calling them a wave goes; Integrated Information Theory is no longer called established; amplification, reinforcement learning from human feedback and Landauer's objection are cited; the square-root cap is defined where it is named; the abstract keeps the author's own voice; two definition titles lose their nested brackets; Section 3.3 states the supercritical regime once; Table 3 says fitted where it said measured; Figure 1's boxes state the bound as registered; Section 5 names Law I's status; Appendix E.2 scopes the Philosophical Statement's differential to the prior documents it surveys; a line before the moved notes says where their corrections stand, and Table 6 maps the layout of version 1.16; the version number and the date. No registered law, prediction, kill condition or dependency changes; no measured number or fitted value changes; no date of any claim moves.

B.2 Earlier versions, as printed

Each entry and note is kept as printed. The positions they name (Section 1.4, Section 3.5 and so on) refer to the version in which they were written; Table 6 gives where each section of version 1.16 now stands.

Version 1.16 (5 October 2026): the title now follows the programme's uniform order (paper, its law or framework, its own title) and reads *Paper I · The ARC Theory · Law I, the ARC Principle · The ARC Equation, the Law of Conversion*; until this version it read *Law I, the ARC Principle: the ARC Equation, the Law of Conversion (Paper I)*. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. Mentions of the programme's papers in this paper now carry their uniform titles. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Version 1.15 (4 October 2026): the title now names the law the paper states, Law I, the ARC Principle, and the title block beneath the author line carries the same title; until this version the paper was titled *The ARC Equation: the Law of Conversion (Paper I)*, and it was first published on 17 January 2026 as *Eastwood's ARC Principle: Preliminary Evidence for Super-Linear Capability Amplification Through Sequential Self-Reference*, the subtitle the title block still carried. The note on how to read this paper now says that it states Law I in formal terms and gives a preliminary result for it, and that its place is the first empirical look at Law I, resting on the Foundational paper. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Version 1.14: The theory gained two laws on 24 September 2026, and this version brings the paper up to date with that and with the programme's record; each sentence corrected in place is recorded, with what it printed before, in the correction note of 3 October 2026 beneath this entry. Substantive: the note on which law this paper carries now says that the theory has five laws, Laws I to III registered on 8 September 2026 and Laws IV and V added on 24 September 2026, neither registered as a law nor tested, with their mathematics set out at version 1.0 of 3 October 2026, and both it and the title note give the ARC Principle as the registered name of the First Law, where they gave it as the name of the three together; a dated note beside Section 6.2 places the paper against Laws IV and V; the epistemic-status line now says that Laws I to III are under registered adversarial test and that Laws IV and V are not yet registered as laws and have not been tested. Wherever the paper printed a figure of Paper II's fit for Gemini 3 Flash, it now carries that model's row whole, as one model's fit and inconclusive, with the bootstrap interval beside the endpoint exponent it belongs to; in row F1, a count of point estimates above 1 gives way to

Paper II's own statement that such estimates come only from ceiling-compressed data. The paragraph on dated predictions, recorded on 25 August 2026, stands as recorded, and a dated note after it gives the state of registration as read on 3 October 2026: the theory-level predictions were registered on 8 September 2026, the corrector-class audit (unit 06) was submitted for registration on 1 October 2026, and every other current trial is a draft. The state of the trials named in row F4 and in Sections 5.2 and 5.3 is brought up to date in place, and Section 5.2 records that Study AE was withdrawn on 12 August 2026. Editorial: two arXiv works in Section 1.4 carry their authors, their titles and their first-posted dates, three cited works there carry their second authors, and the Tononi reference its registered title; one heading in Section 5.1 takes a hyphen in place of a dash; cited works are linked to their official sources where one could be confirmed, and the programme's papers to their DOIs; the version number and the revision date. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Correction note (3 October 2026). Some sentences of this paper had become false through later events, or did not match Paper II, the programme's register or the cited works. This version corrects them in place, and this note records what they printed before. The title note and the note on which law this paper carries spoke of the theory's three laws; the theory has had five since 24 September 2026. Both gave the ARC Principle as the name of the three together, as it was at the renaming of 15 August 2026; the theory registration of 8 September 2026 names the First Law the ARC Principle, and both notes now say so. The epistemic-status line called every Law the programme names a conjecture under registered adversarial test; Laws IV and V are neither registered as laws nor tested, and the line now says so. The abstract, the reading guide, the table of claims in Section 1.3, the caption of Figure 1, the finding in Section 3.4, rows F1 and F4 of Section 4, Sections 5.1 and 5.2 and the conclusion printed the regression exponent of Paper II's fit for one model as a six-model or cross-architecture best fit, as a bare point, or with the bootstrap interval that belongs to that fit's endpoint exponent; row F1 also counted 2 of 6 point estimates above 1, a count that Paper II's table of results by model does not give. Each now gives Paper II's row for Gemini 3 Flash whole. Section 5.3 said that the corrector-class audit and the corrector-class contrast were both drafted and awaiting submission. Row F4 called the artefact-mediated estimation registered. Section 5.2 called the exponent-composition identity registered, in two sentences, and named Study AE, without its withdrawal, as one of the two drafted registrations carrying it. Section 1.4 dated two arXiv works to 2026, though both were first posted in 2025, and gave them by identifier alone, without author or title; it named only the first author of a paper by White and Seymour and of two papers by Kozlowski and Konarzewski. No result is added or refitted, and no date or falsification criterion moves.

Version 1.13: Two dated notes have been added, each beside the passage it governs, and both are additive. Substantive, in Section 2.3: the domain on which the reciprocal equivalence holds is now printed, together with what happens at the value where it breaks down and above it. Substantive, in Section 3.5: the two rows compared there are recorded as differing in system, in evaluation context and in resource axis at once, so that comparison stands as a hypothesis and not as an inference from those two rows. Editorial: the version number and the revision date. No printed sentence has been removed or rewritten, no value refitted, no table changed and no falsification criterion weakened; the two-point figure of 1.34 stands as published, as this paper's original evidence, with the qualifications already printed beside it.

Version 1.12: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

B.3 Notes moved from the body, as printed

These notes stood beside text that version 1.17 corrects; they are kept as printed, and the entry for version 1.17 (Appendix B.1) gives what changed, the benchmark of the sequential pair among it.

Domain note (19 September 2026). An owner review of the equivalence above finds it exact on the domain this model admits and false outside it, and this note prints the domain that the sentence leaves implicit. For $\gamma < 1$ the quantity $1 - \gamma$ is positive, so $1/(1-\gamma) \leq 2$ multiplies to $1 \leq 2(1-\gamma)$, that is $\gamma \leq 1/2$, and the implication reverses by the same step: the equivalence holds in both directions. At $\gamma = 1$ the quantity $1 - \gamma$ is zero, so $1/(1-\gamma)$ is undefined and the reciprocal inequality has no truth value, while $\gamma \leq 1/2$ reads $1 \leq 1/2$ and is defined and false. Exactly one side is undefined, and the displayed equivalence is therefore not defined at that point. For $\gamma > 1$ the quantity $1 - \gamma$ is negative, so $1/(1-\gamma) < 0 < 2$ holds for every such γ while $\gamma \leq 1/2$ holds for none, and the equivalence fails. The domain $\gamma < 1$ is exactly the range on which $\alpha_{\text{crit}} = 1/(1-\gamma)$ is positive and finite and can serve as a ceiling at all, so the restriction is the model's own rather than an assumption added to it, and it is what the clauses above already imply where they state that the relation is derived inside a pacing model and is offered as a minimal-model result rather than as a general law. Read inside that domain, the point of the sentence stands entire: the ARC Ceiling and the half-value are one commitment written in two notations, and neither may be offered as evidence for the other. Unchanged: the law-before-value statement; the single assumption named as producing the value two; the minimal-model status; and the consequence that if that value dies the law survives, because γ is measured and the ceiling it implies is reported.

Comparison note (19 September 2026). An owner review of this comparison finds that the two rows it rests on differ in more than the form of recursion, and this note states what is held fixed across them and what is not. The parallel row comes from a system whose resource variable is the number of samples under majority voting, reported in September 2024; the sequential row comes from a different system, built by a different developer, whose resource variable is a thinking-token count, reported four months later. The benchmark is common to both. The system under test, the evaluation context and the quantity on the horizontal axis vary together across the pair. A difference in fitted exponent between them is consistent with a role for the form of recursion, and it is equally consistent with those other differences, so the sentence above is to be read as a hypothesis rather than as an inference from the two rows. The comparison that would isolate the form of recursion holds the system and the resource definition fixed and varies only the form, and Section 4 is where its outcome is recorded, at F1 and F4. Unchanged: both tables, both fitted values, the direction of the difference, and the falsification criteria.

This paper and Laws IV and V, 3 October 2026. The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law is registered as a law, neither has been tested, and their mathematics is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. This paper states Law I and tests it preliminarily on published data from one mathematics benchmark, AIME 2024. Its own argument about values is the conditional passage above, on embedded values and early alignment, and the paper measures nothing about values. Two other passages could be mistaken for tests of Law V, and neither is one. The contrast between sequential and parallel recursion compares two forms of recursion on capability, and the comparison note in Section 3.5 records that its two rows also differ in the system under test, the evaluation context and the quantity on the horizontal axis, so it stands as a hypothesis. The prediction in Section 5.3 about human-in-the-loop oversight concerns the class of the corrector, not where correction is placed. The paper does not test whether a gain installed by external post-training persists under later capability-directed

training, whether build order moves the correction-leverage exponent, or whether correction that takes part in each revision round pulls away from correction that sees only the finished output as depth rises. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes no result printed here: the results, their dates and their limits stand as they were.

B.4 Earlier titles

Title note. Until 15 August 2026 this paper was published as "The ARC Principle: Formalisation and Preliminary Validation of Recursive Capability Scaling". At that renaming the ARC Principle named the three laws of the theory together, as it then stood; since the theory registration of 8 September 2026 it is the registered name of the First Law, which this paper presents: the ARC Equation, the law of conversion. Nothing in the body was changed by the renaming. (The note as printed in version 1.16. The 1.15 and 1.16 entries above give the titles of 4 and 5 October 2026 and the title of first publication, 17 January 2026; OSF records the paper as *Paper I · The ARC Theory · Law I, the ARC Principle · The ARC Equation, the Law of Conversion*.)

B.5 Where the layout of version 1.16 now stands

Table 6. Where each part of the version 1.16 layout stands in this edition.

Part of version 1.16	Where it stands in this edition
Title note	Appendix B.4
Which law this paper carries	Appendix E.1
How to read this paper	The abstract, Section 4.4 and Appendix E.2
Sections 1 and 1.1, the introduction and its background	Section 1
Section 1.2, the ARC Principle	Section 3.1; the letters' earlier senses in Appendix D.2
Section 1.3, scope and claims	Section 6.1
Section 1.4, related work in context	Section 2; its record of dated predictions in Appendix C.2
Sections 2, 2.1 and 2.2, the theoretical framework	Sections 3 and 3.2
Section 2.3, the quadratic stability ceiling hypothesis	Section 3.3, with Proposition 1 and its proof in Appendix A; the domain note in Appendix B.3
Sections 3 and 3.1 to 3.4, the empirical analysis	Sections 4 and 4.1 to 4.4
Section 3.5, the summary of findings	Section 4.5; the comparison note in Appendix B.3
Section 4, the falsification criteria	Sections 5 and 5.1
Section 5, the limitations	Section 7
Section 5.1, what would refute the principle	Sections 5.2 to 5.4
Section 5.2, the exponent-composition identity	Section 6.2
Section 5.3, the prediction about human-in-the-loop oversight	Section 5.5
Sections 6 and 6.1 to 6.3, the implications	Section 8; the note on Laws IV and V in Appendix B.3

Part of version 1.16	Where it stands in this edition
Section 7, the conclusion	Section 9; its correction note of 3 October 2026 beneath the 1.14 entry, Appendix B.2
Acknowledgments	The Acknowledgements, which hold the author's disclosure alone; the authorship sentence in the copyright block
References	The references, listed in the contents
Reproducibility	Code and data; the standing covenant at the end of Section 5; the epistemic-status line replaced by the status paragraph of Section 6.3; the version entries in Appendix B.2

C Priority record

C.1 Dated records

Table 7 gives each dated record that bears on this paper with its kind of timestamp; no kind is presented as stronger than it is. Version dates of this paper are in Appendix B.

Table 7. Dated records bearing on this paper, each with its kind of timestamp.

Date	Record	Identifier	Kind of timestamp
8 December 2024	The principle stated in an earlier form, as the ARC Hypothesis, in a self-emailed manuscript bundle	10.17605/OSF.IO/QASRK (manuscripts, 5 October 2026; headers, 15 October 2026)	A message the author sent himself, dated by the signed header time of the mail servers that carried it, which shows it existed by that date and is not a sealed timestamp
2 January 2026	The principle in print, with $U = I \times R^2$: *Infinite Architects*, ISBN 978-1806056200 (print; ebook 6 January)	Eastwood (2026a)	Publication date (the author's record)
17 January 2026	This paper first published, stating Law I formally as $U = I \times R^{\alpha}$; the programme's OSF project created	10.17605/OSF.IO/6C5XB	OSF server dates (the project, 08:17 UTC; this paper's upload, 09:18:20 UTC); the paper's own date of first publication
17 March 2026	This paper's own OSF component created and its PDF deposited	10.17605/OSF.IO/B6N27	OSF server date (01:16 and 01:17 UTC)
12 August 2026	Study AE withdrawn before filing; Study AG and the artefact-mediated estimation written for this paper (unit 24) preregistered by timestamp, among the first sixty-five designs	Bitcoin block 962066	Bitcoin anchor (AG, unit 24); the withdrawal is dated inside the programme's record
15 August 2026	Renamed from its earlier title (Appendix B.4)	This paper	Date written in the paper's own record
25 August 2026	The record of dated predictions for this paper (Appendix C.2)	10.17605/OSF.IO/C3NTF	Date written in the paper's own record
28 August 2026	The corrector-class contrast (unit 01) preregistered by timestamp	Bitcoin block 964465	Bitcoin anchor

Date	Record	Identifier	Kind of timestamp
8 September 2026	Laws I to III and the twenty-two propositions registered (version 1.100; updated 13 September 2026, version 1.102)	10.17605/OSF.IO/P8CKQ	Registry record (OSF)
24 September 2026	Laws IV and V added to the theory	None	Date inside the document (the author's record)
1 October 2026	Unit 06, the corrector-class audit, registered; public on 4 October 2026	10.17605/OSF.IO/RV2B5	Registry record (OSF)
6 October 2026	Unit 06 run under its registration; reported in full, administratively incomplete, with no inferential verdict	10.17605/OSF.IO/4BPHG	Date inside the results report; the report deposited on OSF on 11 October 2026, server-dated
15 October 2026	Laws IV and V registered as laws (version 1.103)	10.17605/OSF.IO/P8CKQ	Registry record (OSF)
22 October 2026	Unit 83, the counted test of P21, the placement limb of Law V, registered; public on 25 October 2026	10.17605/OSF.IO/MN73W	Registry record (OSF)
27 October 2026	Unit 82, the counted test of P17, the premise beneath Law III's value two, registered; public the same day	10.17605/OSF.IO/JUFG4	Registry record (OSF)
27 and 28 October 2026	Unit 83 run under its registration; its result is given in Section 6.3	10.17605/OSF.IO/98HKR	Dates written in the run's own records; results deposited on OSF on 29 October 2026, server-dated
29 October 2026	The theory registration updated (version 1.104), adding no prediction and changing no law's statement	10.17605/OSF.IO/P8CKQ	Registry record (OSF)

C.2 Dated predictions and this paper, as recorded on 25 August 2026

One artefact bearing on this paper is catalogued row by row in the programme's machine register, and it is the weakest kind: arc_principle_results.json embeds its falsification thresholds in the file, but carries no timestamp or dated header inside the JSON body, so its ordering cannot be established from the artefact itself. The register grades it weak for that reason and says so in the row. Each row names its file in the public repository with its date basis, so any reader can check the ordering without trusting this page. The programme's full dated chain, from the sealed [Table 7 gives this record's kind of timestamp, which is not a sealed one] manuscript bundle of 8 December 2024 through the printed prediction appendices of 2 January 2026 and the March 2026 preregistration folder to the standing unproven wagers, is assembled in the dated predictions register, together with its machine-readable twin. Forward statements and retrospective matches are never summed. "Registered" is used only for an accepted registry submission, and that submission click remains outstanding across the programme; "preregistered in substance" is used, always with its qualifier and always beside its attestation class, for a prediction whose text was public and timestamped before its outcome existed, as the twelve-row manifest committed at 00:19 UTC on 17 March 2026 was, twenty-three minutes before its fits.

Note (3 October 2026). The paragraph above was recorded on 25 August 2026. Since then the theory-level predictions have been registered on OSF, on 8 September 2026 at version 1.100, and the theory registration was updated to version 1.102 on 13 September 2026. One study unit, the corrector-class audit (unit 06), was submitted for registration on OSF on 1 October 2026; when read on 3 October 2026 its approval period at OSF was still running and it was not yet public. Every other trial is a draft awaiting the author's submission, most of them preregistered by timestamp. Unchanged: the paragraph above and every date in it.

D Notation and glossary

D.1 Symbols

Table 8. Symbols used in this paper.

Symbol	Meaning	Where defined
U	Effective capability on a specified metric	Section 3.1; metric bridge in Section 4.2
I	Base intelligence (single-pass processing capacity without recursive reasoning)	Section 3.1
R	Recursive depth (number of self-referential processing iterations)	Section 3.1
α	Capability scaling exponent; the growth-model value in Figure 1 is asymptotic for $\beta_L < 1$	Section 3.1; Figure 1
α_{err}	Error-decay exponent; subscripts seq and par specify the sequential or parallel series	Equation (3)
R_{res}	Measured resource: sample count or average thinking tokens	Section 4.2
C_{err}	Reciprocal-error proxy, $1/E$, defined only for $E > 0$	Section 4.2
Error_1, Error_2	Positive error rates at resource levels $R_{\text{res},1}$ and $R_{\text{res},2}$	Equation (3)
γ	The correction-leverage exponent	Section 3.3
α_{crit}	The ceiling of stable self-improvement, $1/(1-\gamma)$ (Law III)	Equation (2)
β_L	The self-referential coupling, in $\alpha = 1/(1-\beta_L)$	Figure 1; Section 5.2
β_C, k	Law II's correction-strength exponent and the drift exponent it must exceed	Section 3.3
$g(R)$	Recursive amplification in Figure 1, $[1 + (a/\alpha)R]^{\alpha}$, with $a > 0$ a rate constant and R counting the passes after the first	Figure 1
d	The dimension of a physical distribution network, in $\alpha = d/(d+1)$	Figure 1

D.2 Earlier senses of two letters

U**.** Era note: the letter U carried other senses in earlier layers of this programme, from Universe in the December 2024 formalism to realised creation in a later statement of the law; only the operational sense is

used here.

Terminology note on alpha. Alpha appears in this programme in two distinct meanings across eras: the fine-structure constant in the December 2024 formalism, and the ARC Bound exponent in the 2026 work. In the current analysis, α denotes capability growth and α_{err} denotes error decay; Appendix F records the distinction added in this edition.

D.3 Glossary

Parallel recursion Definition 2.

Sequential recursion Definition 3.

Test-time compute Computation a model spends at inference to reason before responding (Section 1).

The ARC Principle, the ARC Equation Law I of the ARC Theory and its formula, $U = I \times R^{\alpha}$; "the Law of Conversion" is the author's dated name for the ARC Equation, not another law.

The ARC Ceiling, the ARC Bound Law III, $\alpha_{\text{crit}} = 1/(1-\gamma)$, and the claim that this ceiling is at most two for correctors of the system's own class, its point value two a conjecture (Section 3.3).

A law under test A law of the ARC Theory before its registered verdict.

Preregistered by timestamp A design whose text was fixed and timestamped before its data, anchored as Appendix C records; "registered" is kept for an accepted registry submission.

E The paper's place in the ARC Theory

E.1 The five laws and this paper

Which law this paper carries. The ARC Theory has five numbered laws. Laws I to III were registered on 8 September 2026. Laws IV and V, the ARC Persistence Law and the ARC Embedding Law, were added to the theory on 24 September 2026 and registered as laws on 15 October 2026 (version 1.103 of the theory registration); each is a law under test, Law IV not yet tested and the registered result on Law V's placement limb given in Section 6.3, and their mathematics is in development (Laws IV and V: the mathematics, version 1.2, 29 October 2026). The theory registration names the First Law the ARC Principle; the ARC Equation is its formula. This paper carries the First Law, the ARC Principle: $U = I \times R^{\alpha}$, recursion converting into capability at a measured exponent. Paper X carries the Second Law, the ARC Co-Scaling Law. The statement paper carries the Third Law, the ARC Ceiling. The Foundational paper is the cross-domain foundation of Law I and the formalism of the growth exponent that Law III caps.

The laws as registered. As version 1.104 of the registration states them: Law I, the ARC Principle, $U = I \times R^{\alpha}$: usable capability is a power of recursive depth, with the exponent measured rather than assumed. Law II, the ARC Co-Scaling Law, $\beta_C > k$: correction has the asymptotic scaling advantage where its exponent exceeds the drift exponent, burden has it where the correction exponent is the smaller, and at equality coefficients, delay, backlog and saturation decide the outcome. Law III, the ARC Ceiling, $\alpha_{\text{crit}} = 1/(1-\gamma)$: the growth exponent is capped by the reciprocal of the correction shortfall; it is a conditional specialisation of Law II, and the programme claims the inequality, a ceiling of at most two for correctors of the system's own class, and conjectures the equality. Law IV has two limbs: its retention limb is P15's registered claim, word for word, and its comparison limb has no registered proposition yet. Law V carries P12, on build order, and P21, on where correction is placed.

E.2 Its position

This paper's slot inside the theory is the first empirical look at Law I, built on the Foundational paper. Its differential against prior work is in Section 2; the differential of the doctrine the Eden Protocol rests on, against the prior documents it surveys, is in section II.A.8 of the Philosophical Statement.

F Metric clarification of 1 November 2026 and preserved wording

F.1 Version 1.18: what changed and why

Version 1.17 was revised on 31 October 2026; the paper was first published on 17 January 2026. Version 1.18, revised on 1 November 2026, distinguishes error decay against a resource coordinate from capability growth against recursive depth. The numerical observations and their reported uncertainty are unchanged. The correction follows equations (1) and (2) and the reported results of Paper II, cited version 2.19, revised 29 October 2026, and entries EQ-01 and EQ-07 of the Operational Definitions register, cited version 2.0.0, revised 1 November 2026.

For a positive-error power model on the same resource axis, reciprocal error has the same numerical exponent. Identifying it with the capability parameter requires the additional metric, depth and growth-model premises set out in the main text. Algebraic inversion of that parameter supplies no independent coupling measurement. The criteria on capability remain; their status no longer treats an error fit as a deciding capability measurement. The one-half comparison incorporates the correction already stated in Paper II on 19 September 2026. No new experiment, analysis of study data or numerical result is reported.

Appendix B remains the preceding historical record, with its dates fixed. The excerpts below preserve every replaced substantive phrase, paragraph, equation and table cell from version 1.17. An excerpt is the precise changed span, so some are fragments of longer passages retained in the main text. In table excerpts, vertical separators mark the original cell boundaries. These are historical statements, superseded where the correction applies, and must not be read as this edition's current conclusions. The complete earlier figure follows with its earlier wording. The added bibliography entries identify the editions read; all earlier entries are retained.

F.2 Earlier wording, grouped in the correction order

Earlier wording M01 (04 analysis).

> To determine α, we use the power-law relationship. For bounded accuracy metrics, we analyse error rate reduction, taking the error rate to fall as a power of depth, Error $\propto R^{-\alpha}$, so that equation (3) is its slope between two depths on logarithmic axes:

Earlier wording M02 (04 analysis).

$$\alpha = -\frac{\ln(\text{Error}_2/\text{Error}_1)}{\ln(R_2/R_1)}$$

Earlier wording M04 (04 analysis).

Samples (R) | Accuracy (%) | Error Rate (%)

Earlier wording M05 (04 analysis).

Thinking Tokens (R) | Accuracy (%) | Error Rate (%)

Earlier wording M06 (04 analysis).

> **Finding.** Parallel recursion yields $\alpha \approx 0.1$ to 0.3 (sub-linear). Each additional sample contributes less than the previous one.

Earlier wording M07 (04 analysis).

> **Finding.** Sequential recursion yields $\alpha \approx 1.34$ (super-linear) in this two-point estimate. Depth across those two points varies by roughly a factor of 1.9, below the span of at least one and a half decades of recursive depth, a factor of about thirty, that version 1.104 of the theory registration requires of every exponent measurement, which is one more reason to hold the figure as history and not as a value. The paper read this as each additional layer of reasoning amplifying previous gains; since the two points are two versions of the model (Section 4.1), that reading is a hypothesis. The later cross-architecture measurement, on five frontier models, finds architecture-dependent behaviour (Paper II); on Gemini 3 Flash, its one interpretable fit, it reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of −1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a parallel exponent of 0.31 by endpoint estimate (the two arms not matched in compute), which is one model's fit and inconclusive; this early estimate is retained as the paper's original evidence, not as the programme's current value.

Earlier wording M08 (04 analysis).

> Fitted exponents by method and form of recursion, by equation (3).

Earlier wording M09 (04 analysis).

> Method | Recursion Type | Fitted α | Classification

Earlier wording M10 (04 analysis).

> **Key finding, held as a hypothesis.** The scaling exponent depends critically on the form of recursion.

Earlier wording M11 (03 principle).

> The architecture of the ARC Principle. $U = I \times g(R)$ with $\alpha = 1/(1 - \beta_L)$, in which β_L denotes the self-referential coupling; it is not the correction exponent β_C carried by Law II. In $g(R) = [1 + (a/\alpha)R]^{\alpha}$, $a > 0$ is a rate constant and R counts the passes after the first: g solves $dg/dR = a\, g^{\beta_L}$ with $g = 1$ before any added pass, the growth equation of the Foundational paper, which writes that count $r = R - 1$; equation (1) counts the first pass as $R = 1$, and at large depth both grow as R^{α}. Three regimes: sequential ($\alpha_{\text{seq}} \approx 0.49$ on Gemini 3 Flash, Paper II's regression exponent, with a standard error of 0.20, one model's fit and inconclusive; this paper's earlier two-point estimate was ~1.34, Section 4.4), parallel ($\alpha_{\text{par}} \approx 0.1$ to 0.3 in Section 4.3), and the ARC Bound (at most two for correctors of the system's own class, as registered, its point value two conjectured). Cauchy Unification (Paper VII, a classification argued under its stated axioms): $\alpha = d/(d + 1)$ for physical networks of dimension d, a formula earlier work derived (among them West et al. 1997),

and $\alpha = 1/(1 - \beta_L)$ for intelligence. The earlier $\alpha \approx 2.24$ is retracted: it came from the DeepSeek R1 run of 21 January 2026, the first phase of Paper II (Paper IX, Section 7). Thirteen falsification criteria stand across the Foundational paper; this paper's own four are in Table 4. Sources: this paper and Papers II (the cross-architecture study), III, VII and IX.

Earlier wording M13 (06 scope and status).

So the identity is stated here as a claim preregistered by timestamp for testing, its deciding measurements drafted and dated; if the channels disagree, the unification is withdrawn from every surface it has reached, with the same prominence as the framing. The identity is never called established, and the agreement of an asserted one half with a measured 0.49 (Paper II's regression exponent for Gemini 3 Flash, with a standard error of 0.20, one model's fit and inconclusive, its row given whole in Section 4.4) is never presented as evidence, because the two may share an accumulation law and agreement between sharers is not corroboration.

Earlier wording M14 (00 abstract).

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. Does intelligence compound when each step builds on the last, or merely accumulate? This paper formalises and preliminarily tests Law I of the ARC Theory, the ARC Principle (Artificial Recursive Creation), which I first stated in an earlier form, as the ARC Hypothesis, in my manuscripts of 8 December 2024 and published, with the exponent two, in *Infinite Architects* (2026). As registered, it says that usable capability is a power of recursive depth, $U = I \times R^{\alpha}$, with the exponent α measured rather than assumed. The paper's hypothesis is that sequential recursion, in which each step builds on the last, scales super-linearly and faster than parallel recursion over independent samples. On published reasoning-model data, parallel sampling gives $\alpha \approx 0.1$ to 0.3, and two points of sequential reasoning give $\alpha \approx 1.34$; the two series differ in system, benchmark and resource axis as well as in the form of recursion, and the sequential points are two versions of one model, so their contrast is held as a hypothesis. A later measurement on five frontier models (Paper II) finds one interpretable sequential fit, on Gemini 3 Flash: a regression exponent of 0.49 (standard error 0.20) and an endpoint exponent of 0.59 with a bootstrap 95 per cent interval of −1.3 to 2.9. That is one model's fit and inconclusive, so super-linear scaling is unresolved rather than established. The paper also states the ARC Bound, the conjectured value two of Law III's ceiling: $\alpha = 2$ as a stability ceiling, not an impossibility limit, which a durably stable system above two would refute. Falsification criteria and their three-outcome decision form are given. Law I remains a law under test.

Earlier wording M15 (00 abstract).

Version 1.17 states Law I as the theory registration now states it, corrects the sources of the two published series and of the related work, and states the programme's registered results; Appendix B lists every version and correction, with its date.

Earlier wording M18 (00 meta).

it states Law I, the ARC Principle, formally as the ARC Equation and gives a preliminary result for it,

Earlier wording M19 (00 meta).

Appendix B lists every version and correction.

Earlier wording M20 (01 introduction).

Published data from reasoning models suggest that reasoning built step by step gains more from extra effort than voting over independent attempts, but the two series compared differ in more than the form of reasoning, and the one later measurement able to read an exponent cleanly is inconclusive.

Earlier wording M21 (01 introduction).

This paper states Law I formally and gives a preliminary result for it,

Earlier wording M22 (01 introduction).

An estimator of α from error rates at two depths, equation (3), applied to published reasoning-model data: $\alpha \approx 0.1$ to 0.3 for parallel sampling and $\alpha \approx 1.34$ from two points of sequential reasoning, with the reasons their contrast is held as a hypothesis (Section 4).

Earlier wording M23 (01 introduction).

Appendix E the paper's place in the ARC Theory.

Earlier wording M24 (02 related work).

What this paper adds is the statement of the contrast as a difference between exponents of one power law in recursive depth, equation (1), the two-point estimates of Section 4, and the stability ceiling of Section 3.3.

Earlier wording M25 (03 principle).

U = Effective capability (measurable via benchmark performance; the letter's earlier senses in this programme are in Appendix D.2)

Earlier wording M26 (04 analysis).

DeepSeek-R1 | Sequential | ~1.34 | Super-linear (from two points alone; Paper II supersedes it)

Earlier wording M27 (05 falsification).

The ARC Principle (Law I, like every law of the theory a law under test) would be significantly weakened or refuted if:

Earlier wording M28 (05 falsification).

F1 | Sequential recursive depth consistently yields $\alpha \leq 1$ | Under test: Paper II's one interpretable fit, on Gemini 3 Flash (regression exponent 0.49, standard error 0.20; endpoint exponent 0.59, interval −1.3 to 2.9), is one model's fit and inconclusive (Section 4.4). Paper II obtains estimates above 1.0 only from ceiling-compressed data and says they may be inflated. So "consistently" is decided in neither direction (Paper II)

Earlier wording M29 (05 falsification).

> F2 | α decreases as recursive architectures mature | Not met, and not decided: the one result that bears on it, Paper II's cross-architecture study, finds the trend ambiguous, since its more capable models reached the ceiling and a less capable one the floor, leaving only Gemini 3 Flash in measurable range

Earlier wording M30 (05 falsification).

> F4 | More extensive datasets show $\alpha < 1$ for sequential reasoning | Partially met: on Gemini 3 Flash, Paper II's regression exponent is 0.49 (standard error 0.20) and its endpoint exponent 0.59 (bootstrap 95 per cent interval −1.3 to 2.9). Both sequential point estimates sit below 1 and the bootstrap interval spans both regimes; it is one model's fit and inconclusive. The decisive form awaits the artefact-mediated estimation written for this paper (unit 24), a draft preregistered by timestamp (first anchored on 12 August 2026 at block 962066) and awaiting submission by the author

Earlier wording M31 (05 falsification).

> **Exponent-composition mismatch.** If the measured exponent systematically failed to match the predicted $\alpha = 1/(1 - \beta_L)$ from the self-referential coupling β_L (a quantity distinct from the correction exponent β_C that Law II carries), the mechanistic identity, not merely the fitted curve, is falsified.

Earlier wording M32 (05 falsification).

> **The zero-parameter null.** On Gemini 3 Flash, Paper II's regression exponent is 0.49 (standard error 0.20), one model's fit and inconclusive (Section 4.4 gives the row whole). The regression exponent of 0.49 sits almost exactly on a prediction that needs no framework: independent-sample averaging gives 0.5 by the central limit theorem. This is the most serious objection to the empirical case and it is stated in full, with its consequence, as Objection 5 in Section 5.4 of Paper II and as kill condition FALS-022 in the programme's falsification record. It is not repeated here.

Earlier wording M33 (06 scope and status).

> Parallel recursion yields $\alpha < 1$ in AI benchmarks | PRELIMINARY | Limited data (o1)

Earlier wording M34 (06 scope and status).

> Sequential recursion yields $\alpha > 1$ in AI benchmarks | UNRESOLVED | Two points only, from two versions of DeepSeek-R1 (Section 4.1), since displaced by Paper II's cross-architecture study on five frontier models, whose one interpretable fit, on Gemini 3 Flash (regression exponent 0.49, standard error 0.20; endpoint exponent 0.59, bootstrap 95 per cent interval −1.3 to 2.9), is one model's fit and inconclusive (Section 4.4 gives its row whole)

Earlier wording M35 (06 scope and status).

> That $\alpha = 1.3$ is definitively established (more data points are needed)

Earlier wording M37 (09 conclusion).

Parallel recursion yields $\alpha \approx 0.1$ to 0.3 (sub-linear, diminishing returns)

Earlier wording M38 (09 conclusion).

Sequential recursion yields $\alpha \approx 1.34$ (super-linear on these two points alone, from two versions of one model; Paper II's cross-architecture study displaces it and leaves the regime unresolved: its one interpretable fit, on Gemini 3 Flash, is one model's fit and inconclusive, Section 4.4)

Earlier wording M39 (09 conclusion).

In plain terms, the principle claims:

Earlier wording M40 (D notation).

U | Effective capability (measurable via benchmark performance) | Section 3.1

Earlier wording M41 (D notation).

R | Recursive depth (number of self-referential processing iterations); in Section 4, samples or average thinking tokens | Section 3.1

Earlier wording M42 (D notation).

α | Scaling exponent (empirically determined; hypothesised stability ceiling two) | Section 3.1

Earlier wording M43 (D notation).

α_{seq}, α_{par} | The exponent under sequential and under parallel recursion | Section 4

Earlier wording M44 (D notation).

Error_1, Error_2 | Error rates at depths R_1 and R_2 | Equation (3)

Earlier wording M45 (D notation).

Every use of alpha in this paper is the 2026 sense above.

Earlier wording M50 (05 falsification).

Paper II's review of that objection, of 19 September 2026, finds that the half exponent does not follow from independent averaging for the proportion of incorrect responses it fits, so one half is a reference value until a sampling, aggregation and decision model connects it to that outcome, and closeness to it identifies no mechanism; the objection is sharpened rather than removed.

F.3 Earlier figure, retained as historical evidence

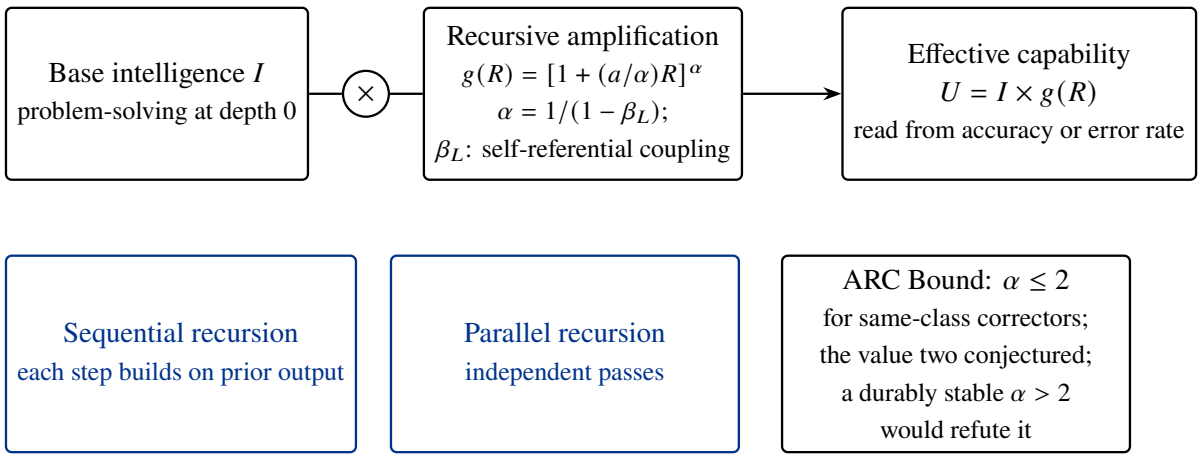


Figure 2. The version 1.17 architecture figure, retained unchanged from 31 October 2026. Its instruction to read capability from accuracy or error rate is superseded by the explicit metric requirement in Figure 1 and Section 4.2. This is a historical reproduction, not a second current model.